

React

```
<BrowserRouter>
  <Header />
  <Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
    <Route path="/product/:id" element={<Product />} />
  </Routes>

  <footer>
    <h1>Footer</h1>
  </footer>
</BrowserRouter>
```

```
import React from "react";
import { useParams } from "react-router-dom";

const Product = () => {
  const { id } = useParams();
  return (
    <div>
      <h1>{id}</h1>
      <h1>Product</h1>
    </div>
  );
};

export default Product;
```

```
import React from "react";
import { Link } from "react-router-dom";

const Header = () => {
  return (
    <div>
      <Link to="/">Home</Link>
      <Link to="/about">About</Link>
      <Link to="/product">Product</Link>
    </div>
  );
};

export default Header;
```

```
import React from "react";

const Home = () => {
  return (
    <div>
      <h1>Home</h1>
    </div>
  );
};

export default Home;
```

```
import React from "react";

const About = () => {
  return (
    <div>
      <h1>About</h1>
    </div>
  );
};

export default About;
```

```
import React from "react";
import Axios from "../Axios";

const Conditional = () => {
  const isLoggedIn = true;
  const userType = "admin";

  switch (userType) {
    case "admin":
      return <h1>Admin dashboard</h1>;
    case "user":
      return <h1>User dashboard</h1>;
    default:
      return <h1>Welcome to the app</h1>;
  }

  // if (isLoggedIn) {
  //   return <Axios/>;
  // } else {
  //   return <h1>Logged out</h1>;
  // }
}
```

```
// return <div>{isLoggedIn ? <h1>Logged IN</h1> : <h1>Logged  
out</h1>}</div>;  
  
// return <div> {isLoggedIn && <h1>Welcome back </h1>}</div>;  
};  
  
export default Conditional;
```

Theory

- **React Router**
 - BrowserRouter → wraps the app to enable routing.
 - Routes → container for all route definitions.
 - Route → maps a path to a component.
 - Link → navigates between routes without reloading.
 - useParams → extracts dynamic route parameters (e.g., /product/:id).
- **Conditional Rendering**
 - **if-else** → render different components based on conditions.
 - **ternary operator** → condition ? <ComponentA/> : <ComponentB/>.
 - **logical &&** → condition && <Component/>.
 - **switch-case** → handle multiple conditions cleanly.

Coding Round Questions

- React Router setup → Create a React app with routes /home, /about, and /product/:id.
- Dynamic route → Implement a product page that displays the product ID using useParams.
- Navigation links → Build a header with Link components to navigate between pages.
- Conditional rendering → Write a component that shows different dashboards based on userType (admin, user, or guest).
- Ternary operator → Use a ternary operator to show “Logged In” or “Logged Out” based on a boolean state.
- Logical && → Render a welcome message only if isLoggedIn is true.

Machine Coding Round Challenge

Problem: Role-Based Dashboard with Routing

Requirements:

1. Set up routes:
 - / → Home page.
 - /about → About page.
 - /product/:id → Product details page.
2. Create a Header with navigation links.
3. Implement a Conditional component that:
 - Shows **Admin Dashboard** if userType = "admin".
 - Shows **User Dashboard** if userType = "user".
 - Shows **Welcome Page** otherwise.
4. Add login state (isLoggedIn) and render either:
 - <h1>Logged In</h1> if true.
 - <h1>Logged Out</h1> if false.
5. Use useParams to display product ID dynamically on the product page.

Expected Flow:

- User clicks **Product** → navigates to /product/101 → sees product ID displayed.
- If userType = "admin" → Admin dashboard is shown.
- If isLoggedIn = false → "Logged Out" message is displayed.